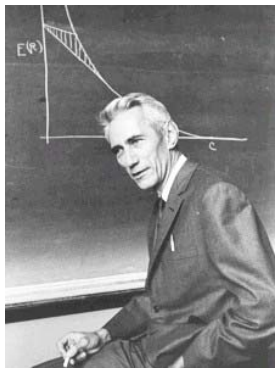
Reliable Computation From Unreliable Components

Theory for Classical Computers



Claude Shannon
(1948)

- Error correcting codes

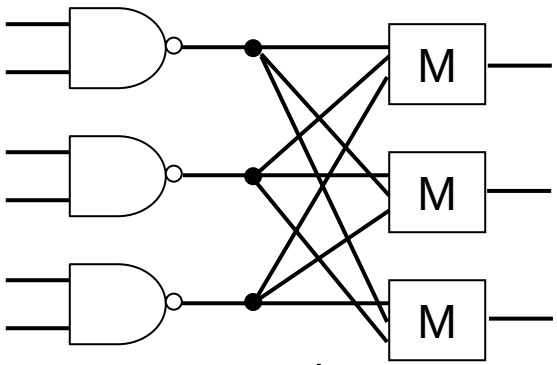
$[7, 4, 3]$ Hamming code

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$



John von Neumann
(1956)

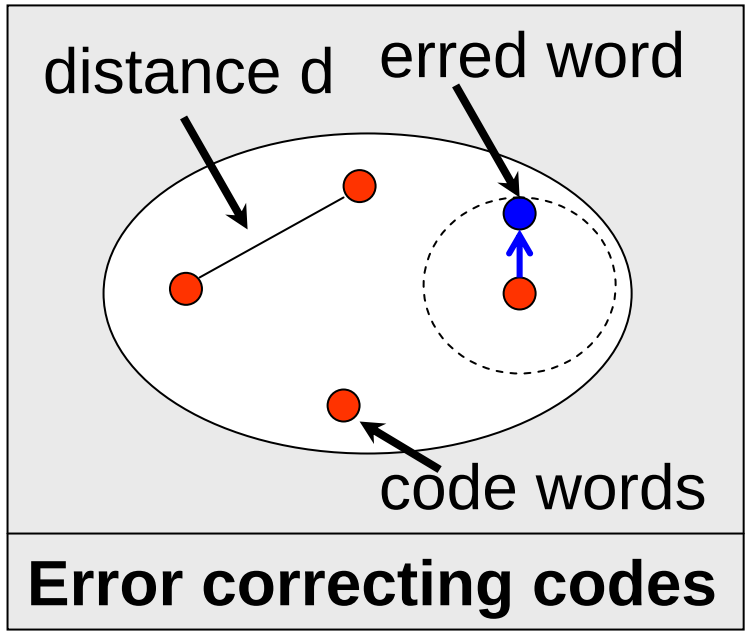
- Fault-tolerance



error rate = p

$\cong$

error rate = $O(p^2)$



Error correcting codes

Reliable Computation From Unreliable Components

Practice for Classical Computers



1st transistor



1st integrated circuit



Pentium chip



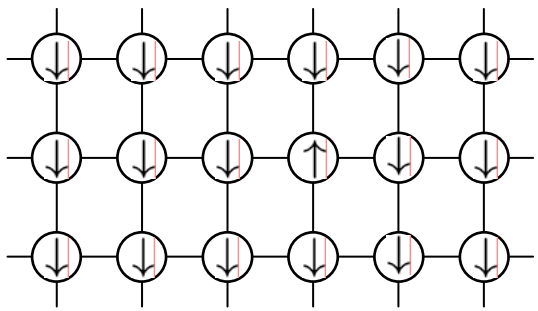
5.25” floppy



Hard drive

Theory concepts implemented by **the physics** of these devices

Example:
magnetic media



- redundancy code
- physical dynamics enacts error correction
- “self-correcting”

Reliable Computation From Unreliable Components

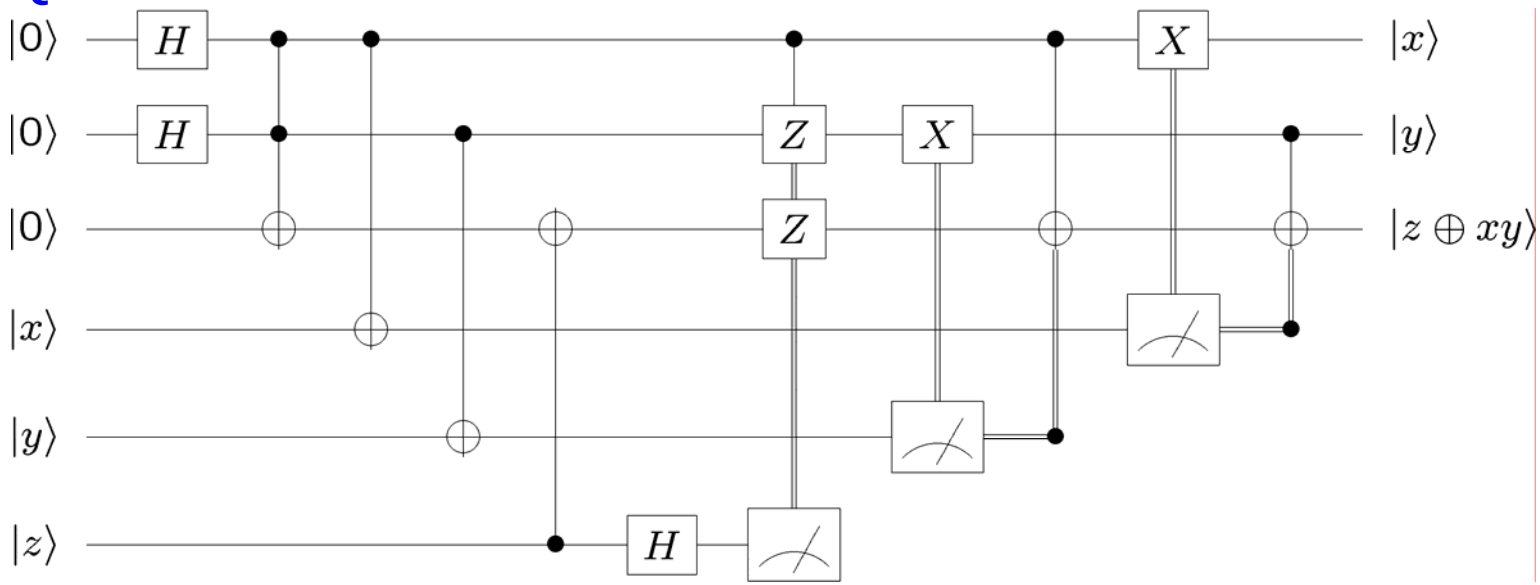
Theory for Quantum Computers

- Quantum error correcting codes

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$

$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

- Quantum fault-tolerance



Reliable Computation From Unreliable Components

Practice for Quantum Computers

Standard path — Quest for scalable architecture operating below the threshold for fault-tolerant quantum computation

Drawbacks: large device complexity,
reduced threshold for realistic architectures, …

A different path (Kitaev 97)

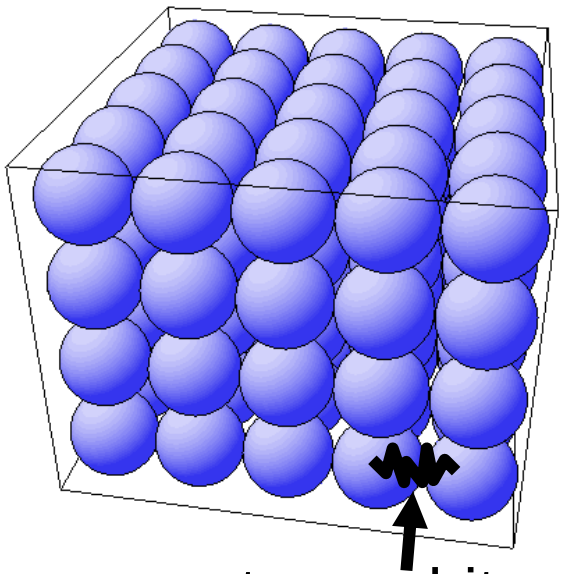


Quantum hard drive?

Do there exist (or can we engineer) materials whose **physics** guarantees fault-tolerant quantum computation?

Self-Correcting Quantum Computers

Strongly interacting many-qubit system



two qubit interactions

1. Information encoded into a quantum error correcting order parameter

2. Quantum error cause string-like excitations with energy proportional to perimeter

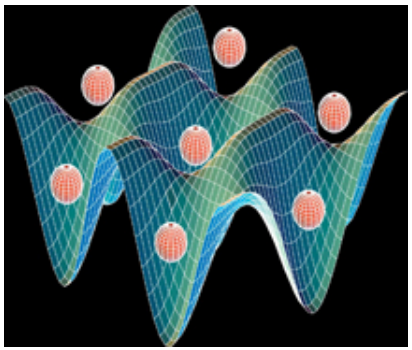
3. Below critical temperature system is self-correcting

Challenges

New codes: quantum error correcting subsystems
New physics: quantum error correcting order parameters
New architectures: cellular approaches to error correcting micro-architecture

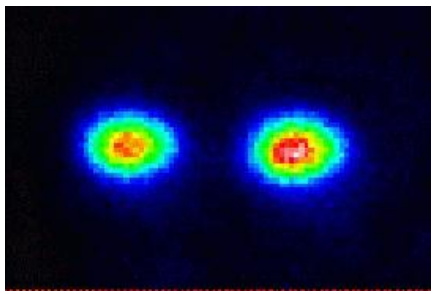
Self-Correcting Quantum Computers

Physical implementation in



optical lattices
(picture from NIST)

Neutral atoms trapped in crystalline potential created by standing waves of light



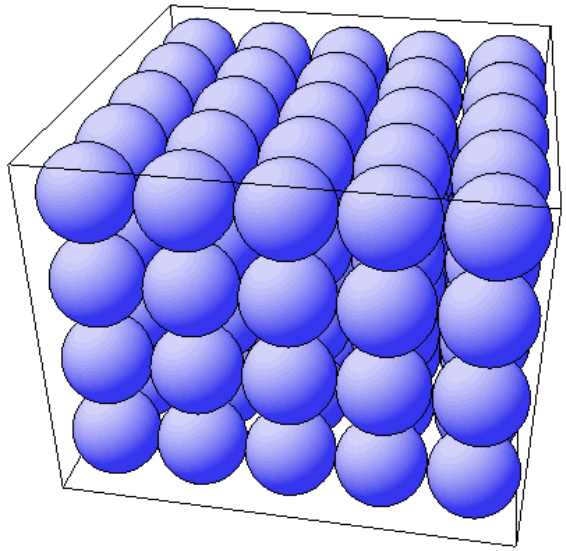
novel ions traps
(picture of ions from NIST)

Two and three dimensional ion traps with hundreds of ions

Self-Correcting Quantum Computers: The Road Ahead

Expertise

Theory of quantum codes, strongly interacting quantum systems, quantum fault tolerance….



Experiment

Experimental challenges of engineering strongly interacting systems…

Engineering

Bridging divide of theory and experiment for realistic optimal device design…